

Task 1 – Thermal Model Assumptions and Calibration

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Objective

The objective of Task 1 was to develop a reduced-order thermal model of the NovaCool data center capable of estimating rack inlet and outlet temperatures from workload power consumption. The model was designed to capture first-order thermal behavior while remaining computationally efficient for control and reinforcement learning applications.

1. Modeling Assumptions

Assumption 1: Reduced-Order Thermal Model

The facility was represented using a lumped-parameter thermal model rather than Computational Fluid Dynamics (CFD).

Each rack row was treated as a single thermal zone characterized by:

- Average cold-aisle temperature
- Average hot-aisle temperature

This assumption was consistent with the project requirement to use a simplified thermal model.

Assumption 2: Facility Layout

The NovaCool facility consists of:

- 200 racks
- 4 rows
- 50 racks per row

Rows were mapped as:

- Row 1 → Racks 0–49

- Row 2 → Racks 50–99
 - Row 3 → Racks 100–149
 - Row 4 → Racks 150–199
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Assumption 3: One CRAH Unit per Row

The facility contains four CRAH units.

A one-to-one mapping was assumed:

- CRAH 1 → Row 1
- CRAH 2 → Row 2
- CRAH 3 → Row 3
- CRAH 4 → Row 4

This assumption was necessary because detailed airflow routing information was not provided.

Assumption 4: IT Power is Converted to Heat

Nearly all rack electrical power consumption was assumed to be converted into heat.

Therefore: $Q_{\text{heat}} \approx P_{\text{IT}}$

This is a standard assumption in data center thermal modeling.

Assumption 5: Rack Outlet Temperature Calculation

Rack outlet temperatures were calculated using the energy balance equation:

$Q = \dot{m}C_p\Delta T$ which was rearranged as:

$\Delta T = Q / (\dot{m}C_p)$ where:

- Q = rack heat load
 - $\dot{m}$ = airflow rate
 - C_p = specific heat of air
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Assumption 6: Uniform Airflow Distribution

Airflow supplied by each CRAH was distributed equally among the 50 racks within its row.

Therefore: Airflow per rack = Row airflow / 50

This assumption was required because rack-level airflow measurements were unavailable.

Assumption 7: Constant Thermophysical Properties

The following engineering constants were treated as constant:

- Specific heat of air: 1005 J/kg-K
- Specific heat of water: 4186 J/kg-K

Property variation over the operating temperature range was considered negligible.

Assumption 8: Ambient Leakage

Hot-aisle temperatures were modified using:

$$T_{\text{hot}} = 0.95 \times T_{\text{hot}} + 0.05 \times T_{\text{ambient}}$$

This represents imperfect hot-aisle/cold-aisle containment and allows limited interaction between the data hall and ambient environment.

Assumption 9: Ambient Temperature Profile

The project specification provided a seasonal outdoor temperature range of 15°C–40°C but did not provide a day-specific ambient profile.

Therefore, a representative daily ambient temperature profile ranging approximately from 18°C to 33°C was generated to emulate daytime heating and nighttime cooling.

2. Calibration Process

Initial Conditions

The initial model used:

- Cold-aisle temperature = 18°C
- Hot-aisle temperature = 25°C
- CRAH supply temperature = 18°C
- CRAH airflow rate = 200 kg/s

These values were selected as preliminary engineering estimates.

Validation Results

Comparison against sensor_reference.csv revealed significant disagreement:

- Mean inlet temperature $\approx 9^{\circ}\text{C}$
- Mean outlet temperature $\approx 26^{\circ}\text{C}$

The original assumptions resulted in inlet temperatures that were too high and outlet temperatures that were significantly underpredicted.

Calibration Actions

The following parameters were calibrated using sensor_reference.csv:

1. Cold-aisle initialization
2. Hot-aisle initialization
3. CRAH supply temperature
4. CRAH airflow rate

Calibration focused on improving agreement between simulated and measured outlet temperatures while maintaining physically realistic behavior.

3. Final Calibrated Parameters

Parameter	Final Value	Basis
Cold-aisle temperature	9°C	Mean inlet temperature
Hot-aisle temperature	26°C	Mean outlet temperature

CRAH supply temperature	9°C	Mean inlet temperature
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CRAH airflow rate	80 kg/s per CRAH	Validation against outlet temperatures
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4. Lessons Learned

The largest source of model error was not the governing thermal equations but the initial operating assumptions.

Sensor-based calibration significantly improved model accuracy by grounding the thermal model in measured facility behavior.

The final model should therefore be interpreted as a calibrated engineering approximation suitable for thermal analysis, control design, and RL environment development rather than a high-fidelity CFD representation.

Task 2 – Power and PUE Model Assumptions

Objective

The objective of Task 2 was to estimate facility power consumption and Power Usage Effectiveness (PUE) using the thermal model developed in Task 1.

The power model accounts for:

- IT equipment power
- CRAH fan power
- CRAH pump power

- UPS losses
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Assumption 1: IT Power

Total IT power was calculated as the sum of rack power across all 200 racks.

IT power is treated as the primary useful load of the facility.

Assumption 2: UPS Efficiency

UPS efficiency was modeled as a load-dependent curve rather than a constant value.

The model assumes:

- Lower efficiency at very low utilization
- Peak efficiency near nominal operating load
- Reduced efficiency as load approaches system limits

This behavior is consistent with real UPS systems, which typically achieve maximum efficiency near their design operating point.

Reasoning:

Detailed manufacturer performance curves were not provided, therefore a simplified efficiency model was used to capture realistic efficiency variation with load.

Assumption 3: CRAH Fan Power

CRAH fan power was estimated using the HVAC affinity law: $P_{\text{fan}} \propto \text{airflow}^3$

Implemented as: $P_{\text{fan}} = P_{\text{nominal}} \times (\text{flow} / \text{flow}_{\text{nominal}})^3$

Reasoning:

Fan power increases nonlinearly with airflow and the cubic affinity law is commonly used for HVAC performance estimation.

Assumption 4: CRAH Pump Power

Pump power was assumed to scale linearly with chilled-water flow rate:

$$P_{\text{pump}} = k \times \text{flow_rate}$$

Reasoning: Detailed pump performance curves were not available in the project specification. A linear relationship provides a reasonable first-order approximation for facility-level analysis.

Assumption 5: Cooling Infrastructure Power

Total cooling infrastructure power was calculated as:

$$\text{Cooling Power} = \Sigma(\text{Fan Power}) + \Sigma(\text{Pump Power})$$

Only CRAH fan and pump energy were included in cooling overhead calculations.

Reasoning: Detailed chiller, cooling tower, and auxiliary plant information were not provided.

Assumption 6: Power Usage Effectiveness (PUE)

PUE was calculated using the standard industry definition:

$$\text{PUE} = \text{Total Facility Power} / \text{IT Power}$$

where:

$$\text{Total Facility Power} = \text{IT Power} + \text{UPS Losses} + \text{Cooling Infrastructure Power}$$

Reasoning: PUE is the most widely used metric for evaluating data center energy efficiency. Lower PUE values indicate more efficient facility operation.

Limitations

The power model focuses on first-order facility power accounting and does not explicitly model:

- Chiller power consumption
- Cooling tower power

- Transformer losses
- Lighting loads
- Miscellaneous building loads

Therefore, the reported PUE values should be interpreted as relative efficiency indicators rather than exact facility measurements.

Task 3 – Validation Assumptions and Results

Objective

The objective of Task 3 was to evaluate how closely the thermal model reproduced the behavior observed in the provided sensor reference data.

Assumption 1: Sensor Reference as Ground Truth

The file `sensor_reference.csv` was treated as the reference representation of facility behavior.

Validation metrics were computed by comparing simulated temperatures against the corresponding sensor-derived temperatures.

Assumption 2: Facility-Level Validation

Validation was performed using aggregated facility metrics rather than individual rack comparisons.

The following quantities were evaluated:

- Mean inlet temperature
- Mean outlet temperature
- Peak outlet temperature

Reasoning: The thermal model is a reduced-order facility model and is therefore intended to reproduce overall thermal behavior rather than rack-level temperature variations.

Assumption 3: Calibration Variables

Parameters that were not directly specified in the case study were treated as calibration variables.

These included:

- CRAH supply temperature
- Airflow rate
- Initial thermal conditions

Calibration was performed using sensor measurements while maintaining physically reasonable operating values.

Validation Metrics

The following error metrics were used:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Maximum Absolute Error
- Bias

These metrics quantify both average prediction accuracy and systematic overprediction or underprediction.

Part B – Control and RL Environment Assumptions

Task 4 – Control Policy Assumptions

Objective

The objective of Task 4 was to evaluate whether the calibrated digital twin could be used to operate the cooling system more efficiently while respecting thermal safety constraints.

The primary safety constraint was:

Rack outlet temperature $\leq 40^{\circ}\text{C}$

The secondary objective was to reduce cooling energy and improve PUE.

Assumption 1: Fixed Workload Trace

The workload trace was treated as fixed for all policies. Therefore, all control policies served the same total IT demand.

Reasoning: The task allowed workload redistribution as an optional stretch goal, but this implementation focused on CRAH control only.

Implication: Because workload was not shed or redistributed, policy comparison was based on:

- Thermal violations
- Peak outlet temperature
- Cooling energy
- Mean PUE

rather than differences in total IT load served.

Assumption 2: Control Inputs

The controller was allowed to adjust:

- CRAH supply temperature
- CRAH airflow rate

Workload redistribution was not implemented.

Reasoning: CRAH supply temperature and airflow are direct, interpretable cooling controls and are sufficient to demonstrate the control value of the simulator.

Assumption 3: Baseline Policy

The baseline policy used fixed CRAH settings throughout the 24-hour simulation.

Reasoning: A fixed-setpoint baseline provides a simple reference case for evaluating the benefit of active control.

Assumption 4: Reactive Policy

The reactive controller adjusted cooling based on peak rack outlet temperature.

If peak outlet temperature increased toward the safety limit, the controller increased cooling by:

- Lowering CRAH supply temperature
- Increasing airflow

If the facility was safely below the limit, the controller relaxed cooling to save energy.

Reasoning: This mimics a simple thermostat-style control strategy and is easy to interpret.

Assumption 5: Safety-Aware Optimizer

The safety-aware optimizer used a more aggressive rule-based strategy.

It relaxed cooling when peak outlet temperatures were safely below the limit and increased cooling aggressively when temperatures approached 40°C.

Reasoning: This policy was designed to explore the tradeoff between energy efficiency and thermal safety.

Assumption 6: Control Bounds

Supply temperature and airflow were constrained to remain within physically reasonable operating ranges.

These bounds were centered around the calibrated values from Part A:

- CRAH supply temperature near 9°C
- CRAH airflow near the calibrated row airflow (80)

Reasoning: Without bounds, the controller could choose unrealistic cooling settings.

Task 5 – RL Environment Assumptions

Objective

The objective of Task 5 was to expose the calibrated simulator through a Gym-style reinforcement learning interface.

The goal was not to train an RL agent, but to design an environment that future agents could interact with.

Assumption 1: Gym-Style Interface

The environment follows the standard structure:

`reset()` → initial observation

`step(action)` → next observation, reward, done flag, and diagnostics

Reasoning: This interface is familiar to RL engineers and makes the simulator reusable for future agent training.

Assumption 2: Observation Space

The observation vector includes:

- Mean cold-aisle temperature
- Mean hot-aisle temperature
- Peak rack outlet temperature
- Normalized IT load
- Normalized ambient temperature
- Time of day encoded using sine and cosine
- CRAH supply temperatures
- CRAH airflow rates

Reasoning: The agent needs to observe both the current facility state and the current control settings in order to make meaningful decisions.

Assumption 3: Action Space

The action vector controls incremental changes to:

- CRAH supply temperature
- CRAH airflow rate

for each of the four CRAH units.

Reasoning: Incremental actions are smoother and more realistic than allowing the agent to directly jump to arbitrary setpoints.

Assumption 4: Reward Function

The reward function balances four objectives:

1. Reward IT load served
2. Penalize high PUE
3. Strongly penalize thermal violations
4. Softly penalize operation near the 40°C limit

Reasoning: A reward based only on energy efficiency could encourage the agent to reduce cooling too aggressively. A reward based only on safety would not encourage energy savings. The combined reward provides a more useful learning signal.

Assumption 5: Thermal Safety Priority

Thermal violations were assigned a large penalty.

Reasoning: The prompt defines 40°C as a hard safety constraint, so unsafe operation should dominate the reward signal.

Assumption 6: Episode Termination

Episodes terminate when:

- The 24-hour workload trace is complete, or
- Cumulative thermal violations exceed a safety threshold

Reasoning: The 24-hour termination matches the provided workload horizon. Early termination prevents unstable or unsafe policies from continuing indefinitely.